Array 30 (2026) 100912

Contents lists available at ScienceDirect

Array

journal homepage: www.elsevier.com/locate/array

RuralEdgeHealth: A sustainable offline mobile edge AI reference architecture for intelligent IoT sensor classification in resource-constrained environments

Abubakar Wakili *, Sara Bakkali

Euromed University of Fez, Fez, Morocco

ARTICLE INFO

Keywords:
Edge AI
Internet of Things (IoT)
Mobile edge computing
On-device inference
Offline edge AI
Resource-constrained devices
Sensor classification
Reference architecture

ABSTRACT

The deployment of intelligent Internet of Things (IoT) systems in rural and resource-constrained environments remains limited by unreliable connectivity, restricted energy availability, and the widespread use of entry-level mobile hardware, conditions affecting more than 2.6 billion people worldwide. Although artificial intelligence (AI) enables automated interpretation of sensor data, most existing IoT–AI architectures rely on persistent cloud connectivity and high-performance devices, which limits their applicability in underserved regions. This paper introduces RuralEdgeHealth, an offline Android mobile edge AI reference architecture designed for on-device IoT sensor classification under strict connectivity and hardware constraints. The proposed framework follows a modular five-layer design consisting of: (1) a presentation layer supporting Bluetooth Low Energy (BLE) sensing and manual input; (2) an IoT sensor integration layer enabling standardized data acquisition; (3) an edge intelligence layer performing fully on-device inference using ONNX-optimized model; (4) an encrypted local data management layer based on SQLCipher; and (5) a runtime optimization layer enabling energy-efficient execution on entry-level Android smartphones. A comparative evaluation of four machine learning algorithms: Random Forest, XGBoost, LightGBM, and Multi-Layer Perceptron, shows that tree-based models provide the most suitable trade-off between accuracy and computational efficiency for deployment-constrained environments. The Random Forest model achieves 98.25% classification accuracy with sub-100 ms inference latency on smartphones with 2 GB RAM. A preliminary exploratory usability assessment also highlights accessibility differences between technical and non-technical users, providing early design insights for inclusive edge AI systems rather than definitive population-level usability conclusions. The results demonstrate the technical feasibility of offline-first mobile edge AI systems and provide a reusable reference architecture for extending intelligent IoT applications to connectivity-constrained environments, supporting the objectives of Sustainable Development Goals 9 and 10. The study is intended as a Phase 1 architectural and deployment feasibility evaluation, with future work focused on real-world data validation, larger usability studies, and field deployment testing

1. Introduction

Internet of Things (IoT) technologies enable distributed sensing and data-driven decision support across domains such as healthcare, agriculture, environmental monitoring, industrial automation, and smart infrastructure. By combining sensor networks with machine learning, IoT systems can convert raw sensor measurements into actionable information that supports monitoring, prediction, and operational decision-making. Despite these capabilities, the deployment of AI-enabled IoT systems in rural and infrastructure-limited environments remains constrained by three interrelated factors: unreliable network connectivity, limited energy availability, and the widespread use of entry-level mobile hardware [1–3].

Fig. 1 summarizes the concept investigated in this study. In many practical deployments, particularly in rural and low-resource regions, conventional IoT–AI architectures assume persistent cloud connectivity and access to high-performance computing resources. These assumptions do not hold in many parts of the world. Recent estimates indicate that more than 2.6 billion people still lack reliable internet access, which significantly restricts their ability to benefit from intelligent sensing and automated data analysis systems [4]. Consequently, architectures that depend heavily on cloud infrastructure often fail to operate effectively in connectivity-constrained environments.

Most existing IoT platforms rely on cloud-centric processing models in which edge devices mainly serve as data collection nodes while data

* Corresponding author.
E-mail address: a.wakili@ueuromed.org (A. Wakili).

https://doi.org/10.1016/j.array.2026.100912
Received 29 June 2025; Received in revised form 12 May 2026; Accepted 14 May 2026
Available online 21 May 2026
2590-0056/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

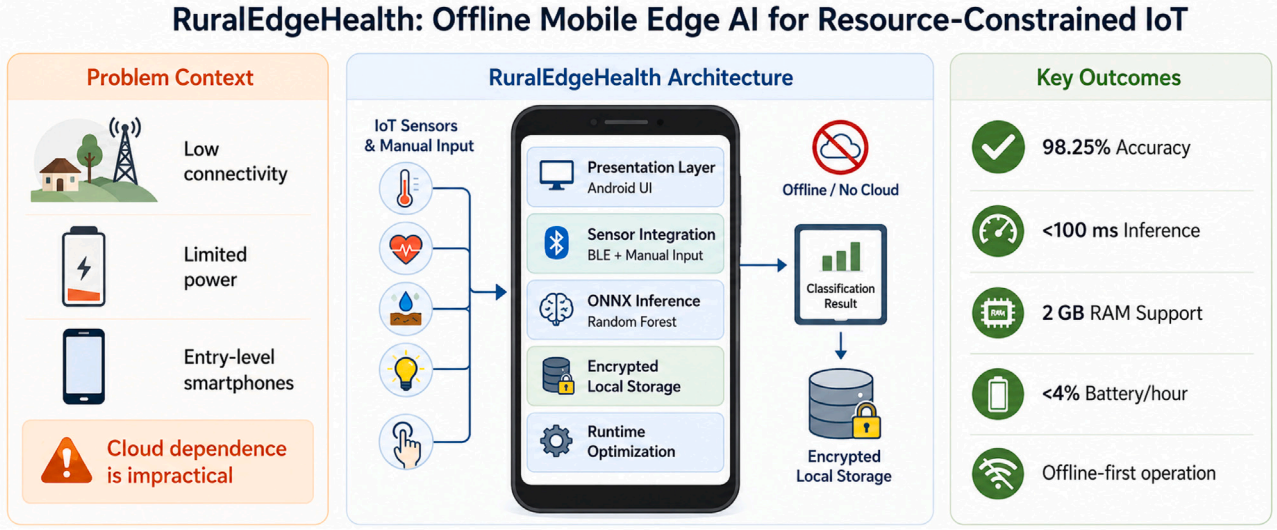


Fig. 1. Graphical abstract illustrating the RuralEdgeHealth framework: an offline-first mobile edge AI reference architecture enabling IoT sensor classification on entry-level smartphones using BLE/manual input, ONNX-based inference, encrypted local storage, and energy-aware runtime optimization for connectivity-constrained environments.

analysis and machine learning inference are performed in centralized cloud systems [5,6]. Although this architecture is suitable for well-connected settings, it introduces several limitations in infrastructure-constrained environments. Continuous network dependence increases latency, reduces system reliability under unstable connectivity conditions, and raises privacy concerns due to persistent data transmission. More importantly, this design reinforces a technological imbalance: communities that could benefit most from accessible sensing technologies often have the least infrastructure available to support conventional AI systems [5,7]. These challenges limit progress toward inclusive digital infrastructure development aligned with Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure) and SDG 10 (Reduced Inequalities) [8,9].

Edge computing offers an alternative approach by allowing machine learning inference to be executed directly on local devices instead of relying entirely on remote cloud services [10,11]. This design reduces dependence on continuous connectivity, shortens response times, and allows data to remain on the device where it is generated [12,13]. However, many existing edge AI systems still present barriers for deployment in low-resource environments. First, several proposed solutions target mid-range or high-end devices that require 4–8 GB of RAM and modern processors, which excludes the entry-level smartphones commonly used in many developing regions [14,15]. Second, some implementations maintain implicit reliance on cloud services for tasks such as model synchronization, configuration management, or telemetry reporting, preventing true offline operation [15–17]. Third, the literature lacks well-defined architectural frameworks that systematically address the combined constraints of limited hardware resources, restricted connectivity, and energy efficiency.

Although individual components relevant to offline IoT systems – such as Bluetooth Low Energy (BLE) sensing, lightweight inference optimization, and local data management – have been investigated in isolation [18,19], recent reviews emphasize the absence of integrated architectural solutions that combine these capabilities into a reusable design for resource-constrained deployments [1,20,21]. As a result, many implementations remain application-specific and difficult to generalize across different IoT domains. This gap highlights the need for a structured reference architecture capable of supporting reliable and efficient AI-enabled IoT operation in environments where connectivity and computational resources are limited.

To address this need, this paper introduces RuralEdgeHealth, an offline mobile edge AI reference architecture designed for IoT sensor classification under strict connectivity and hardware constraints. The proposed framework performs machine learning inference directly on entry-level Android smartphones without requiring internet connectivity or cloud infrastructure. Although the architecture is demonstrated through a health-oriented vital signs classification scenario – chosen because physiological monitoring provides a clear and widely understood sensing context [22] – the framework itself is domain-agnostic and applicable to other IoT applications including environmental sensing, agriculture, and industrial monitoring.

The system adopts a modular five-layer architecture designed to operate efficiently on entry-level Android devices (Android 8.0+, 2 GB RAM). A comparative evaluation of four machine learning algorithms – Random Forest, XGBoost, LightGBM, and Multi-Layer Perceptron – shows that tree-based models are particularly suitable for resource-constrained deployments. Among the evaluated models, Random Forest achieves the best balance between predictive accuracy, computational efficiency, and model size, reaching a classification accuracy of 98.25%, sub-100 ms inference latency, and a compact model size of approximately 3.5 MB.

The novelty of RuralEdgeHealth lies not in proposing a new machine learning algorithm, but in integrating deployment-critical capabilities into a single reusable reference architecture for constrained environments. In contrast to existing edge/offline AI frameworks, RuralEdgeHealth combines complete offline inference, compatibility with entry-level Android smartphones with 2 GB RAM, quantified latency/battery/memory evaluation, manual input fallback when sensors are unavailable, and exploratory multi-group usability assessment. This architecture-level contribution addresses a practical gap between laboratory edge AI prototypes and deployable offline intelligent IoT systems for underserved settings. The main contributions of this work are summarized as follows:

- **A reusable five-layer reference architecture for offline-first mobile edge AI**, providing a structured design framework for deploying intelligent IoT applications on resource-constrained Android smartphones. The architecture integrates BLE-based sensing, manual fallback input, ONNX-optimized on-device inference, encrypted local storage, and energy-aware runtime optimization.
- **A systematic empirical evaluation of lightweight machine learning models** under realistic deployment constraints, demonstrating that tree-based models – particularly Random Forest – provide the most favorable trade-offs among accuracy, latency,

energy consumption, and storage footprint on entry-level mobile devices.

- **Evidence-based insights into accessibility and usability**, derived from a multi-group usability evaluation that highlights differences between technical and non-technical users and informs inclusive design considerations for edge AI systems.
- **Open-source implementation for reproducibility**. The RuralEdgeHealth framework, including implementation code, trained models, and datasets, is released under the MIT License to support transparent and reproducible research.

The remainder of this paper is organized as follows. Section 2 reviews related work on edge AI systems and offline IoT architectures. Section 3 describes the reference architecture and experimental methodology. Section 4 presents the system implementation. Section 5 reports the experimental results, including model performance, cross-device benchmarks, battery consumption, usability evaluation, and task completion analysis. Section 6 provides a comparative analysis with representative edge AI IoT systems. Section 7 discusses the principal findings, implications, threats to validity, and limitations. Section 8 outlines future research directions, and Section 9 concludes the paper.

2. Related work

The development of edge AI and offline-capable IoT systems has attracted growing research interest as connectivity limitations and infrastructure constraints increasingly necessitate local data processing. Despite substantial progress, a critical examination of existing literature reveals persistent gaps between demonstrated technical capabilities and the practical deployment requirements of resource-constrained environments.

2.1. Edge AI applications in IoT systems

Edge computing approaches for IoT applications have demonstrated technical feasibility but frequently fail to address accessibility requirements critical for deployment in resource-constrained settings. Amin and Hossain [10] presented a comprehensive survey of edge intelligence and IoT, analyzing AI-based classification and prediction techniques for sensor data processing. While their study identified key challenges, including data quality, availability, and real-time processing limitations, it lacked empirical implementation and did not address hardware accessibility constraints essential for deployment in low-resource environments.

Recent advances in edge AI optimization have shown promise across multiple application domains. Surianarayanan et al. [23] surveyed optimization techniques for edge AI, identifying quantization, pruning, and model compression as key strategies for resource-constrained deployment. However, their analysis focused primarily on algorithmic efficiency, without considering accessibility or usability constraints relevant to deployments in developing regions.

Pathinarupothi et al. [24] demonstrated an IoT-based smart edge system for remote monitoring, reporting a 95% alert transmission success rate. Despite explicitly targeting deployment in remote and rural developing regions, their implementation relied on hardware specifications beyond those typically available in such environments, thereby undermining practical accessibility.

Izhar et al. [25] proposed IoT–edge fusion approaches incorporating advanced security mechanisms based on distributed ledger technology and elliptic curve cryptography. While technically sophisticated, their framework required mid-range smartphone specifications and maintained cloud dependencies for optimal performance, systematically excluding the populations most in need of accessible IoT solutions.

Although several studies focus on healthcare as a primary application domain, the limitations identified – particularly those related to hardware accessibility and offline capability – extend to intelligent IoT systems across domains such as agriculture, environmental monitoring, and smart infrastructure.

2.2. IoT systems and energy efficiency

Energy efficiency remains a critical challenge for IoT deployment in resource-constrained environments. Talpur et al. [26] investigated energy-efficient monitoring using smartphones and IoT technologies, identifying power management as a primary barrier to sustained operation. Their work emphasized the importance of optimizing both hardware and software components to support extended use in regions with limited charging infrastructure.

Mishra et al. [27] developed a low-cost IoT-based monitoring system focused on real-time data collection and management. Although their approach addressed cost considerations, it retained cloud dependencies that significantly limit deployment feasibility in connectivity-constrained environments.

Tawalbeh et al. [16] proposed an edge-enabled IoT system model emphasizing local data processing for enhanced privacy protection. However, their implementation primarily addressed security concerns, without tackling the practical energy and accessibility challenges that arise in low-resource deployments.

2.3. Offline-first and privacy-preserving IoT systems

Offline-capable IoT systems have emerged as an important research direction for addressing connectivity limitations. Rancea et al. [12] conducted a systematic review of edge computing approaches, highlighting benefits related to privacy, security, and real-time decision-making. Nevertheless, their analysis remained conceptual, offering no empirical validation or deployable architectural frameworks suitable for connectivity-limited environments.

El Zouka and Hosni [28] developed secure IoT communication mechanisms for smart monitoring systems, emphasizing privacy-preserving data exchange. While their work addressed critical security requirements, it did not consider the fundamental accessibility barriers that hinder deployment in underserved regions.

Alzubi et al. [29] presented a comprehensive review of privacy and security challenges in edge computing for smart IoT systems. Although their study identified key vulnerabilities and open research directions, it did not propose practical solutions tailored to the constraints of resource-limited environments, where conventional security mechanisms may be infeasible.

2.4. Systematic reviews and implementation gaps

Several systematic reviews have documented the challenges and opportunities associated with edge IoT systems, while also revealing significant implementation gaps. Hartmann et al. [11] conducted a systematic review identifying requirements and challenges across diverse IoT use cases. While their work effectively catalogued technical barriers, it offered no actionable strategies for overcoming accessibility constraints in underserved regions.

Saif-Ur-Rahman et al. [21] examined the role of artificial intelligence and digital technologies in improving service delivery in low- and middle-income countries (LMICs). Their review highlighted the transformative potential of AI-driven solutions, while also identifying persistent barriers related to infrastructure, training, and technology accessibility.

Okwu et al. [30] conducted a systematic scoping review of artificial intelligence for strengthening systems in LMICs, identifying key challenges such as inadequate infrastructure, limited technical expertise, and insufficient funding. However, their analysis focused primarily on problem identification rather than proposing concrete technical solutions.

2.5. Recent advances in edge computing

Recent developments in edge computing demonstrate increasing algorithmic and architectural sophistication, yet accessibility challenges remain unresolved. Wang et al. [31] proposed an edge AI system integrating generative AI and explainable AI for sustainable edge–cloud computing. Despite its technical advancement, the system required substantial computational resources beyond the capabilities of entry-level devices.

Hennebelle et al. [32] developed a machine learning-based framework for prediction tasks in integrated IoT, edge, and cloud computing environments. While demonstrating technical feasibility, the framework maintained cloud dependencies that limit applicability in connectivity-constrained settings.

Cruz and Bertemes Filho [17] proposed architectural improvements for edge–IoT systems supporting continuous monitoring, emphasizing user-centered design. However, their implementation encountered significant battery life and sensor reliability challenges, without providing effective solutions for the power management constraints critical to rural deployments.

2.6. Critical analysis: The accessibility–sophistication paradox

A synthesis of existing literature reveals a persistent *accessibility–sophistication paradox*: systems with advanced AI capabilities typically require hardware and infrastructure specifications that exclude the very populations they aim to serve, while solutions designed for accessibility often sacrifice analytical sophistication. This paradox manifests through three recurring limitations:

- **Infrastructure Assumptions**: Many implementations assume stable electricity, continuous connectivity, and access to recent devices—conditions that are not met in large portions of rural and underserved regions. Even studies claiming to target global or rural deployment frequently retain these exclusionary assumptions.
- **Performance-First Design Philosophy**: Existing work prioritizes technical performance metrics, such as accuracy and latency, over accessibility considerations. As a result, gains in analytical performance are often achieved at the expense of deployability on accessible hardware.
- **Fragmented Solutions**: Current approaches address individual technical components in isolation, resulting in partial solutions that lack end-to-end integration of edge AI, offline capability, and accessibility-oriented design.

2.7. Research gaps and positioning

The analysis identifies three critical gaps in the current literature. **Hardware Accessibility Gap**: No existing system demonstrates sophisticated AI inference on genuinely accessible hardware; most implementations require mid-range to high-end devices, excluding resource-constrained populations. **Offline AI Sophistication Gap**: While some systems support offline operation, none sustain high-accuracy AI inference without connectivity, with offline-first designs typically trading analytical capability for autonomy. **Comprehensive Integration Gap**: Existing solutions address isolated components rather than providing integrated reference architectures that combine IoT connectivity, advanced edge AI, and accessibility-oriented design.

RuralEdgeHealth addresses these gaps through a design philosophy that treats accessibility as a primary architectural constraint rather than a secondary consideration. Unlike existing work that regards offline operation as a degraded mode, the proposed reference architecture maintains high-accuracy AI performance with full offline functionality on accessible hardware, challenging prevailing assumptions about unavoidable trade-offs between analytical sophistication and accessibility in intelligent IoT systems.

3. Methodology

3.1. Reference architecture overview

RuralEdgeHealth is a reference architecture for *offline* intelligent IoT sensor classification in resource-constrained environments. It follows edge computing principles while explicitly addressing two deployment constraints that are often under-specified in prior work: (i) operation under intermittent or absent connectivity, and (ii) compatibility with widely available, entry-level mobile hardware. The architecture is organized as a modular five-layer hierarchy whose components operate end-to-end without cloud dependencies, as illustrated in Fig. 2. We demonstrate the reference architecture using a health-oriented sensor classification case study.

The reference architecture comprises the following layers.

3.1.1. Layer 1: Presentation and interface layer

The Presentation and Interface Layer provides an interactive user interface with multi-modal input designed for diverse user populations. The interface is implemented in Jetpack Compose using an MVVM pattern to support reactive UI updates. Accessibility-oriented design choices include clear visual feedback, simplified workflows, and reduced reliance on dense text to support users with varying literacy and technical backgrounds. A dual-mode input mechanism enables seamless switching between automated BLE sensor acquisition and manual input fallback, ensuring workflow continuity when sensors are unavailable or connectivity is intermittent. The implementation targets ARMv8-based Android devices (Android 8.0+, 2 GB RAM) to reflect entry-level deployment conditions.

3.1.2. Layer 2: IoT sensor integration layer

The IoT Sensor Integration Layer manages BLE connectivity and acquisition with robust fallback behavior. It supports standardized GATT profiles (including profiles aligned with IEEE 11073 where applicable) through service- and characteristic-based interfaces that can be extended to multiple sensor types using standardized UUID mappings. Connection management implements device discovery, pairing, and reconnection with energy-aware communication settings. When BLE data are unavailable, the layer degrades gracefully to manual entry while preserving the same downstream processing and storage pipeline. BLE connection policies and sampling strategies are configured to support the system-level energy target (i.e., <4% battery consumption per hour under the evaluation workload).

3.1.3. Layer 3: Edge intelligence and processing layer

The Edge Intelligence and Processing Layer executes the on-device inference pipeline optimized for resource-constrained hardware. The processing pipeline performs input validation, normalization, feature computation, and lightweight sensor fusion for heterogeneous IoT data streams. The inference engine runs an ONNX-optimized Random Forest model (3.5 MB), achieving 98.25% accuracy, a weighted F1-score of 0.9825, and sub-100 ms inference latency with an active memory footprint below 200 MB on 2 GB RAM devices. To improve robustness under natural class imbalance, training incorporates SMOTETomek-based balancing and feature engineering aligned with the modeled sensing scenario. The classification service outputs a multi-tier status label and triggers optional, evidence-based recommendations within the application workflow.

3.1.4. Layer 4: Data management and persistence layer

The Data Management and Persistence Layer provides a secure, offline-first persistence stack with encryption and performance-aware caching. Local storage is implemented using Room with SQLCipher (AES-256) to protect sensitive data at rest without requiring network connectivity. A lightweight model repository manages the embedded ONNX model and associated metadata (e.g., versioning and lifecycle

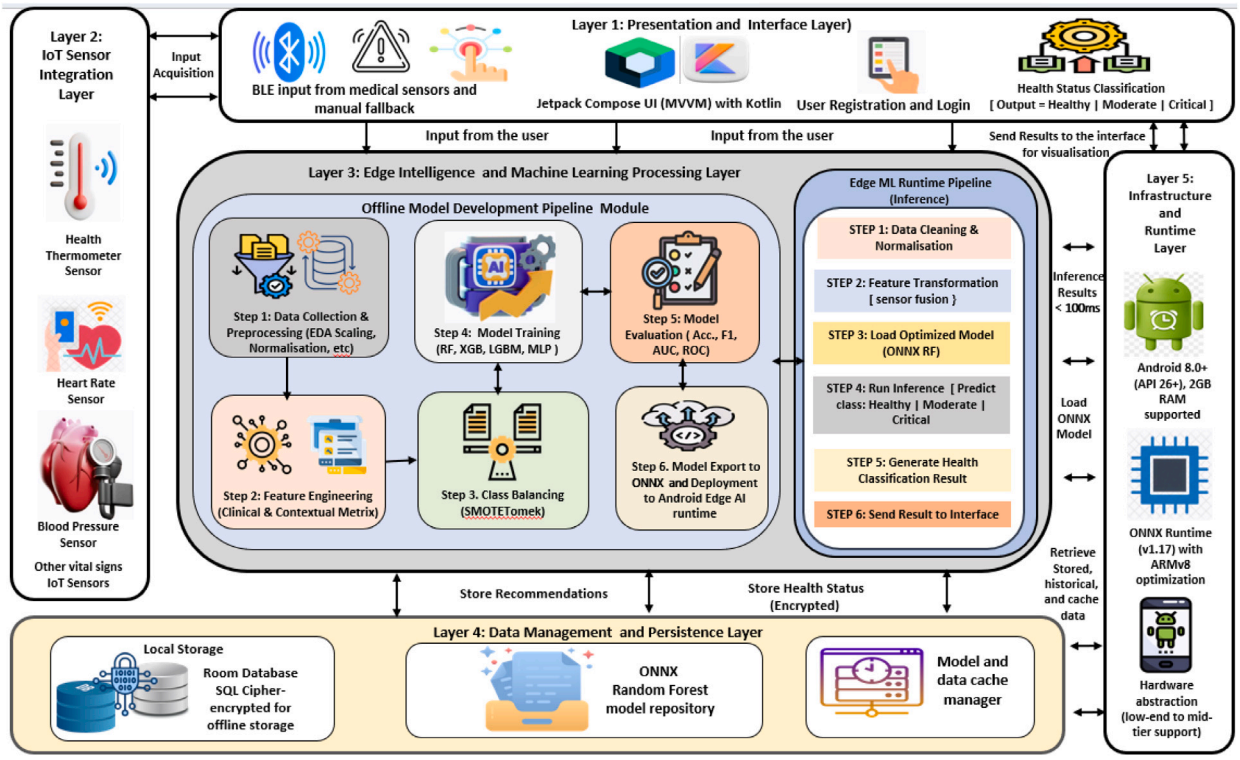


Fig. 2. Proposed RuralEdgeHealth Reference Architecture: A five-layer modular design for offline intelligent IoT sensor classification on resource-constrained mobile devices.

handling). To reduce overhead during repeated queries, the layer implements caching for fast retrieval of historical outputs. Key material is managed through Android Keystore integration, and optional export/backup capabilities are designed to preserve offline-first behavior (i.e., synchronization is not required for core operation).

3.1.5. Layer 5: Infrastructure and runtime layer

The Infrastructure and Runtime Layer optimizes execution across heterogeneous Android hardware profiles. The runtime targets Android API 26+ to support broad device compatibility (the corresponding coverage claim should be supported with an appropriate citation if retained). ONNX Runtime (v1.17.0) is used with ARM CPU execution optimizations suitable for mobile processors. Resource management includes memory-aware allocation strategies, energy-aware scheduling policies, and performance scaling to maintain responsiveness on entry-level devices while exploiting additional capacity on mid-range hardware. These controls support the system-level goals of sub-100 ms inference and sustained operation under battery constraints.

3.2. Dataset development and analysis

To address the limited availability of IoT sensor datasets that combine measurement and contextual features for offline edge deployment, we developed a synthetic dataset representing sensor classification scenarios under resource constraints. The dataset is intended solely for evaluating the computational feasibility of the reference architecture rather than clinical decision support. This choice is consistent with recent work showing that synthetic data can support model development when real-world data collection is limited by practical constraints (e.g., access, cost, or deployment barriers) [33,34]. Synthetic datasets are particularly useful when controlled variation and class coverage are needed to study model behavior under deployment-relevant conditions [35].

The synthetic dataset was designed to approximate deployment-relevant conditions rather than to replicate a fully real-world operational distribution. This approximation is supported by four design choices: preservation of plausible inter-feature relationships among sensor variables, inclusion of contextual and device-level factors, injection of controlled sensor noise, and retention of natural class imbalance prior to balancing. Accordingly, the dataset supports architectural and computational feasibility assessment under constrained conditions, while not replacing validation on real-world sensor streams.

3.2.1. Dataset design and feature engineering

The dataset includes 15 features grouped into four categories: primary sensor readings (HeartRate, Temperature, SystolicBP, DiastolicBP, SpO2), derived metrics (PulsePressure, HealthGap, TargetBP), device metrics (Device_Battery_Level, Battery_Level, BatteryRatio, Sensor_Location), and contextual variables (Gender, Activity_Level, Medication_Status). Each sample is labeled into one of three status classes based on domain-informed criteria: *Normal* (values within expected ranges given context), *Moderate* (one or more values outside expected ranges requiring attention), and *Critical* (multiple values substantially outside expected ranges requiring immediate attention).

Algorithm 1 summarizes the data generation process. We enforced plausible relationships among variables and injected controlled noise to emulate measurement variability typical of consumer-grade sensors outside laboratory conditions. The final dataset contains 10,000 samples with natural class imbalance (Normal: 50%, Moderate: 35%, Critical: 15%), reflecting expected population-level distributions for monitoring scenarios (Fig. 3).

3.2.2. Exploratory data analysis

We performed exploratory data analysis to characterize feature distributions and inform modeling decisions. Fig. 4 reports the distribution of key sensor readings across the three status classes and indicates clear separation patterns consistent with the dataset design assumptions. Correlation analysis (Fig. 5) and pairwise feature relationships

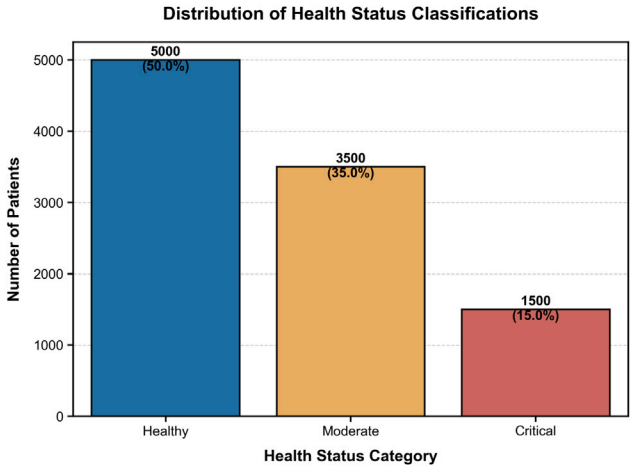


Fig. 3. Distribution of status classifications in the dataset, showing natural class imbalance: Normal (50.0%), Moderate (35.0%), and Critical (15.0%).

Algorithm 1: Domain-Informed Synthetic Data Generation

Input: Target sample size N, feature specifications F
Output: Synthetic dataset D_{synth}

// Step 1: Base Measurements and Demographics
for $i = 1$ **to** N **do**
  Sample: $age \sim \mathcal{N}(45, 15)$, $gender \sim$ Bernoulli(0.5);
  Generate: $\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} \sim \mathcal{N}(\mu_{sensors}, \Sigma_{sensors})$;
end
// Step 2: Derived and Contextual Features
foreach *sample* i **do**
  $DerivedMetric_i \leftarrow f(V_i)$;
  $Device_Battery \sim$ Uniform(20, 100);
  $Sensor_Location \sim$ Cat($[loc_1, loc_2, loc_3]$);
  $Activity_Level \sim$ Cat($[low, mod, high]$);
end
// Step 3: Status Classification
foreach *sample* i **do**
  $score \leftarrow$ ClassificationScore($sensors_i, context_i$);
  if $score < \theta_{normal}$ **then**
    $y_i \leftarrow$ NORMAL;
  else
    if $score < \theta_{critical}$ **then**
      $y_i \leftarrow$ MODERATE;
    else
      $y_i \leftarrow$ CRITICAL;
    end
  end
end
// Step 4: Sensor Noise Simulation
foreach *sensor measurement* **do**
  $sensor_{noisy} \leftarrow sensor + \mathcal{N}(0, \sigma^2_{sensor})$;
end
return $D_{synth} = \{(X_i, y_i)\}_{i=1}^{N}$ *with 15 features per sample*

(Fig. 6) further confirm expected dependencies among measurements and contextual variables, supporting the discriminative capacity of the selected feature set.

3.3. Class imbalance handling

The dataset exhibits natural class imbalance, which can bias training toward majority classes and reduce performance on minority classes

Algorithm 2: SMOTETomek Class Balancing

Input: Dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$, target ratio r
Output: Balanced dataset D'

// Phase 1: SMOTE Oversampling
foreach *minority class* c *with* $|S_c| < r \cdot |S_{maj}|$ **do**
  foreach *sample* $x_i \in S_c$ **do**
    Find k nearest neighbors $N_k(x_i)$ in S_c;
    Randomly select $j \in N_k(x_i)$;
    Generate synthetic sample: $x_{new} = x_i + \lambda(x_j - x_i)$ where $\lambda \sim$ Uniform(0, 1);
    Add (x_{new}, c) to D';
  end
end
// Phase 2: Tomek Links Removal
foreach *pair* (x_i, x_j) *where* $y_i \neq y_j$ **do**
  if $d(x_i, x_j) = \min\{d(x_i, x_k) : x_k \in D', k \neq i\}$ ***and*** $d(x_i, x_j) = \min\{d(x_l, x_j) : x_l \in D', l \neq j\}$ **then**
    Remove both (x_i, y_i) and (x_j, y_j) from D';
  end
end
return D'

that are operationally important. We applied SMOTETomek, which combines SMOTE oversampling with Tomek-links-based undersampling, to improve class balance and boundary cleanliness (Algorithm 2). We used SMOTE with $k = 5$ nearest neighbors and Tomek-links removal with the "all" strategy. Post-processing produced equalized class proportions (33.3% per class). Kolmogorov–Smirnov tests indicated that feature distributions were preserved after balancing ($p > 0.05$), and Fig. 7 illustrates the resulting class distribution shift.

3.4. Model development and selection

We evaluated four candidate algorithms selected to reflect both predictive performance and deployment feasibility on entry-level mobile devices:

- **Random Forest**: robust performance on tabular features, efficient CPU inference, and interpretable feature contributions;
- **XGBoost**: strong gradient-boosting baseline for structured data;
- **LightGBM**: efficiency-oriented gradient boosting for faster training and inference;
- **Multi-Layer Perceptron**: a neural baseline for comparison with classical models.

Rationale. We prioritized classical machine learning models over large deep learning architectures (e.g., CNNs, transformers) due to deployment constraints: (i) model size and memory limits on 2 GB RAM devices; (ii) CPU-only inference latency requirements; (iii) interpretability needs in decision-support settings; and (iv) energy efficiency, as tree-based inference typically requires fewer floating-point operations than neural alternatives.

3.4.1. Random forest implementation

The Random Forest model was trained with the following configuration:

- $B = 100$ trees with bootstrap sampling;
- feature subsampling with $m = \sqrt{15} \approx 4$ features per split;
- majority voting for multi-class prediction;
- 5-fold cross-validation for hyperparameter selection, optimizing weighted F1-score.

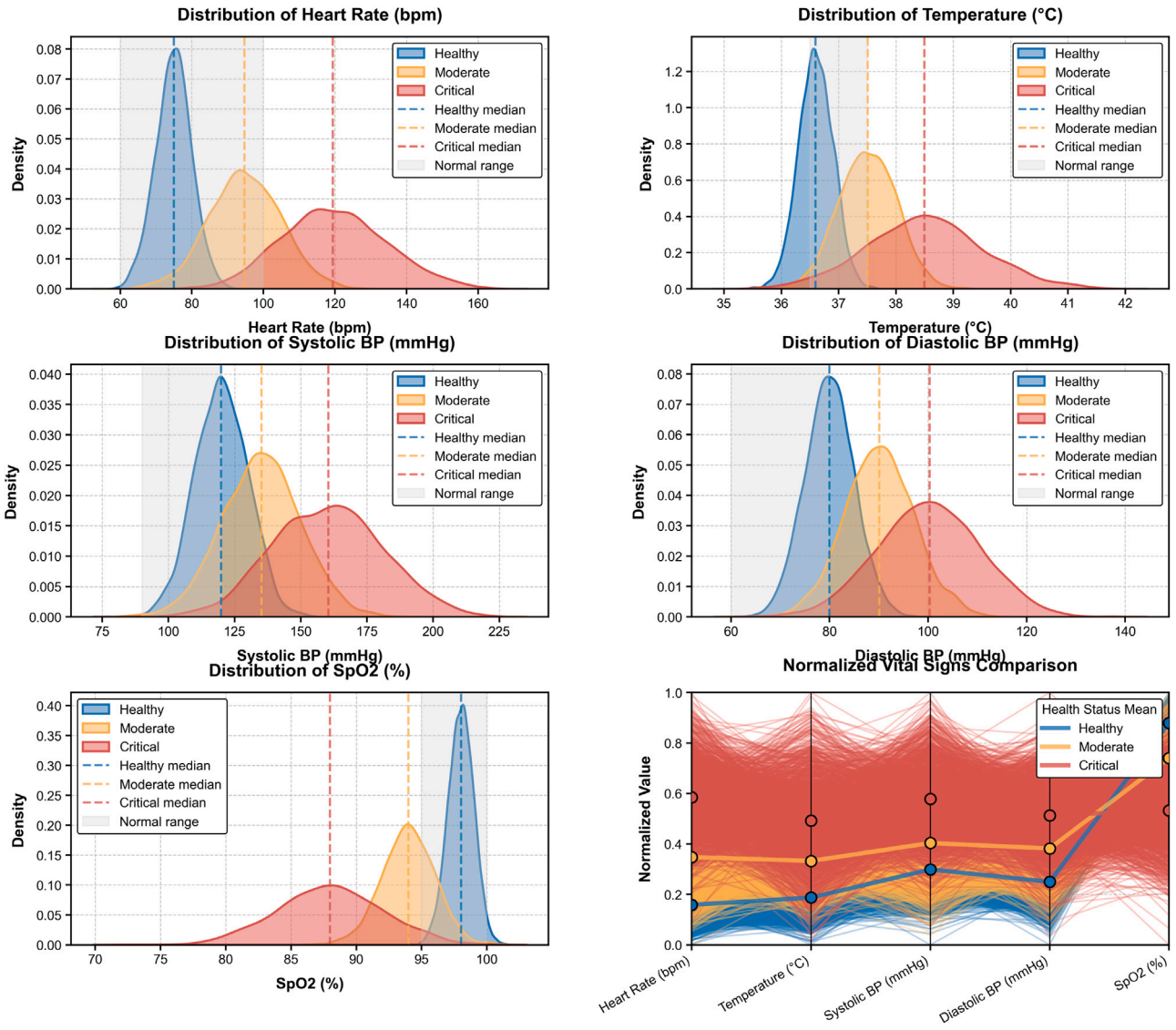


Fig. 4. Distribution of key sensor readings by status category, showing class-dependent patterns consistent with the synthetic data generation process.

3.4.2. Evaluation metrics

We report standard multi-class metrics:

- **Accuracy**: overall correct predictions across all classes;
- **Weighted F1-score**: precision–recall balance weighted by class support;
- **Per-class F1-scores**: class-wise performance for Normal, Moderate, and Critical;
- **ROC–AUC**: one-vs-rest AUC for multi-class discrimination.

3.5. Model optimization for edge deployment

To enable efficient mobile inference, the selected Random Forest model was exported to ONNX and configured for Android runtime execution (Algorithm 3). Conversion was performed using `sklearn-onnx`, producing a portable representation suitable for optimized execution on mobile CPUs. The ONNX model was executed with ONNX Runtime on Android using CPU providers optimized for ARM architectures. We then validated functional equivalence and performance to confirm that accuracy was preserved and that inference latency met the sub-100 ms target on representative devices.

Algorithm 3: Edge Deployment Optimization Pipeline

Input: Trained Random Forest model M, target device specifications
Output: Optimized model M_{opt}

```
// Stage 1: ONNX Model Conversion
M_1 ← ConvertToONNX(M, opset = 11);
// Stage 2: Runtime Configuration
M_opt ← ConfigureONNXRuntime(M_1, providers = [CPU]);
// Stage 3: Model Validation
performance ← ValidateModel(M_opt, D_test);
return M_opt
```

3.6. Evaluation methodology

3.6.1. Technical performance assessment

We evaluated model performance and deployment feasibility using stratified 5-fold cross-validation and the metrics described above. Device-level validation was conducted on representative Android configurations under standardized settings (22 ± 2°C ambient temperature,

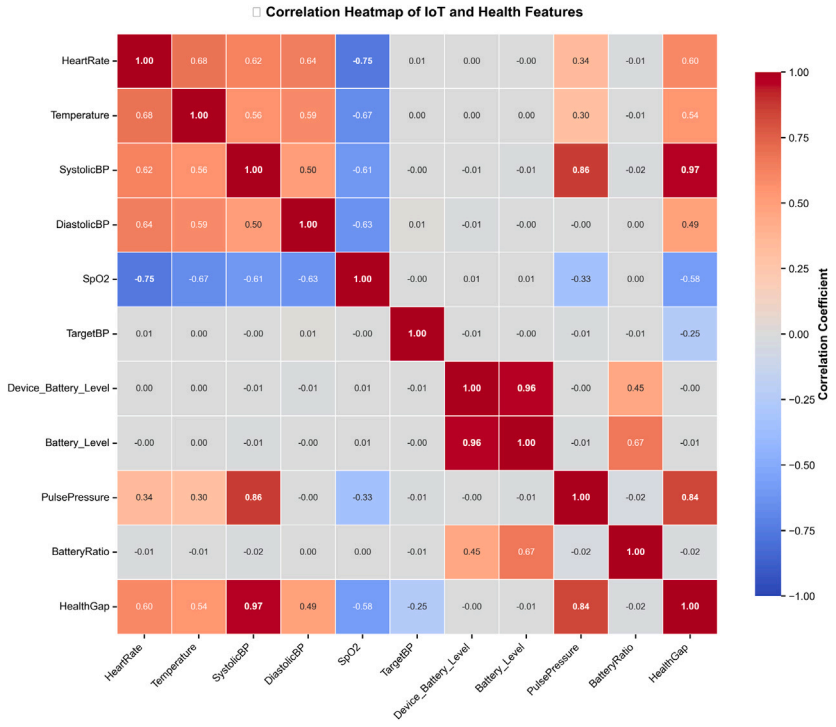


Fig. 5. Correlation heatmap of IoT sensor features, illustrating relationships among measurements and contextual factors.



Fig. 6. Pairwise relationships among key metrics, showing clustering by status category and supporting feature discriminability.

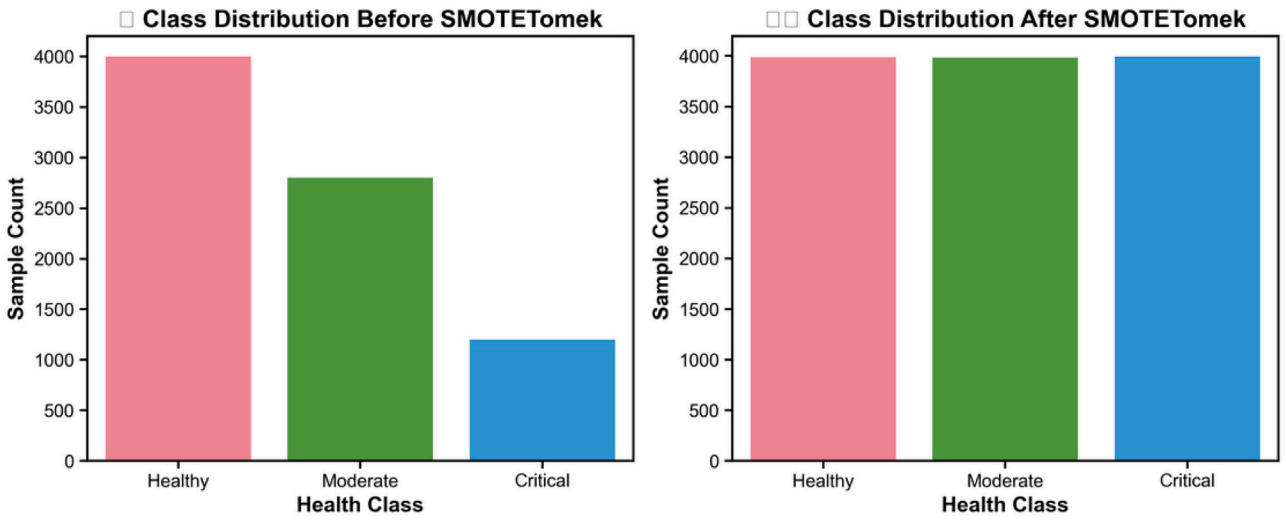


Fig. 7. Class distribution before and after SMOTETomek, showing balancing from 50%/35%/15% to 33.3% per class while preserving feature relationships.

Table 1
Device testing configuration.

Device model	Processor	RAM	Test duration
Android Emulator	x86_64 (AVD)	2 GB	8 h
Samsung Galaxy A10	Exynos 7884	2 GB	8 h
Xiaomi Redmi Note 12S	MediaTek Helio G96	8 GB	8 h

50% screen brightness, airplane mode enabled with Bluetooth active, and background applications cleared) to improve reproducibility.

Controlled experiments were deliberately used in this Phase 1 study to isolate technical feasibility and ensure repeatable measurement of deployment-critical metrics, including inference latency, battery consumption, and memory usage across representative devices. This design enables fair cross-device comparison before introducing the additional variability of field environments, where connectivity instability, device fragmentation, sensor degradation, environmental conditions, and user interaction variability may affect system behavior (see Table 1).

Performance targets were defined to reflect entry-level deployment constraints: inference latency <100 ms, battery consumption <4% per hour under the evaluation workload, accuracy retention >95% after optimization, peak memory usage <200 MB on 2 GB devices, and application package size <50 MB for accessible distribution.

3.6.2. Battery consumption assessment

Battery consumption was measured using the Android `BatteryManager` API following a calibration period to establish baseline device behavior. Battery level was sampled every 15 min under four usage patterns designed to approximate realistic operation: Active Monitoring (25%), Standby Operation (50%), Data Entry Mode (15%), and Export Operations (10%).

3.6.3. System usability scale (SUS) evaluation

Usability was assessed using the standard System Usability Scale (SUS) protocol with stratified sampling across four user groups: technical users ($n = 4$), general users ($n = 4$), semi-technical users ($n = 2$), and low-literacy users ($n = 2$). Sample size was informed by a power analysis ($\alpha = 0.05$, $\beta = 0.20$, $\sigma = 12.5$, $\delta = 10$). We evaluated SUS scores using one-sample t-tests against established interpretive thresholds: acceptable (>68), good (>80), and excellent (>85).

4. Implementation

4.1. Development environment and technology stack

RuralEdgeHealth was implemented using contemporary Android development practices tailored to resource-constrained devices and offline-first operation. Development was carried out in Android Studio Meerkat 2024.3.1 Patch 2, leveraging improved tooling and profiling support for mobile edge AI workloads. We demonstrate the reference architecture using a health-oriented sensor classification case study; however, the software modules are reusable across IoT domains (e.g., agriculture, environmental monitoring, and smart infrastructure).

The technology stack is summarized as follows:

- **Development Platform**: Android Studio Meerkat 2024.3.1 Patch 2 with Gradle 8.5
- **Programming Language**: Kotlin 1.9.22 (with Java interoperability)
- **UI Framework**: Jetpack Compose 1.6.5 for declarative, responsive UI
- **Architecture**: Model–View–ViewModel (MVVM) using Android Jetpack Architecture Components
- **Database**: Room 2.6.1 with SQLCipher for encrypted offline persistence
- **ML Runtime**: ONNX Runtime Mobile 1.17.0 with a custom JNI bridge for on-device inference
- **BLE Integration**: Custom Bluetooth Low Energy stack supporting standardized IoT sensor profiles
- **Target Platform**: Android API Level 26+ (Android 8.0 and above)

4.2. Android application architecture

The RuralEdgeHealth Android application follows the MVVM pattern and Android Architecture Components to ensure clear separation of concerns, robust state management, and resilience under intermittent connectivity. Persistent storage is provided via a Room database integrated with SQLCipher, enabling encrypted local storage suitable for sensitive sensor data and configuration artifacts while maintaining full offline functionality.

4.3. Edge AI model integration

The optimized ONNX model is integrated through an on-device inference pipeline that performs preprocessing, model execution, and

Algorithm 4: Android edge inference implementation.

Input: Raw sensor data $S = \{s_1, s_2, ..., s_{15}\}$
Output: Status prediction $\hat{y}$ and confidence score

```
// Step 1: Model and Parameter Loading
Load ONNX model and preprocessing parameters;
// Step 2: Data Validation
foreach sensor value s_i ∈ S do
    if s_i < min_i or s_i > max_i then
        s_i ← clip(s_i, min_i, max_i);
    end
end
// Step 3: Feature Preprocessing
X_norm ← StandardScaler(S);
X_eng ← FeatureEngineering(X_norm);
// Step 4: Model Inference
P ← ONNXRuntime(M_opt, X_eng);
ŷ ← arg max(P);
conf ← max(P);
// Step 5: Post-processing
if conf < 0.7 then
    ŷ ← UNCERTAIN;
end
Generate guidance based on ŷ;
Cache results with timestamp;
return ŷ, conf, recommendations
```

result interpretation locally. Inference is executed via a custom Java Native Interface (JNI) wrapper over ONNX Runtime Mobile 1.17.0, configured to prioritize low-latency CPU execution on entry-level hardware.

4.4. Mobile application interface

The RuralEdgeHealth application implements a user-centered interface designed to support end-to-end sensor classification workflows while maintaining accessibility for diverse user populations. The interface design emphasizes visual clarity, reduced cognitive load, and progressive disclosure to keep critical functions usable on small screens and low-end devices.

Fig. 8 illustrates the core interface workflow, highlighting authentication, record creation, data acquisition (BLE and manual entry), and longitudinal visualization.

4.5. AI-powered classification and guidance

The application performs AI-driven status classification and presents results through a three-tier framework (Normal, Moderate, Critical) to support rapid interpretation and action. Fig. 9 shows representative result screens across severity levels. The interface combines structured data presentation with a consistent visual hierarchy to support use by both technical and non-technical users.

4.6. Comprehensive records and monitoring

The application maintains comprehensive local records to support both point-in-time assessment and longitudinal monitoring. Fig. 10 presents a representative record view including raw sensor inputs, derived metrics, model outputs, and export utilities for reporting and downstream integration.

Table 2
Comparative performance metrics across model candidates.

Model	Accuracy	F1-score	ROC AUC	Inference time (ms)	Model size (MB)
Random Forest	**0.9825**	**0.9825**	**0.9989**	**87**	3.5
LightGBM	0.9820	0.9820	0.9986	102	4.1
XGBoost	0.9790	0.9790	0.9985	131	4.8
MLP	0.9765	0.9765	0.9981	156	2.2

4.7. Offline-first data management

RuralEdgeHealth adopts an offline-first data architecture using Room with SQLCipher encryption. Sensor readings, derived features, inference outputs, and application configurations are stored locally to ensure uninterrupted operation without connectivity. When connectivity becomes available, the system can synchronize selected artifacts (e.g., exports or backups) without violating offline-first execution constraints.

4.8. Bluetooth low energy sensor integration

BLE connectivity supports standardized IoT sensor profiles and implements energy-aware communication strategies, including connection interval tuning, selective service discovery, automatic disconnection during idle periods, and graceful fallback to manual data entry when sensors are unavailable. This design maintains workflow continuity while reducing power draw during prolonged offline deployment.

4.9. Security and privacy implementation

Security and privacy protections are implemented across storage, access control, and data handling:

- AES-256 encryption for local storage via SQLCipher
- Secure key management using Android Keystore
- Local authentication and role-based access control
- Data anonymization options for research and analytics exports

4.10. Accessibility and cross-cultural design

The interface incorporates design principles intended to support use in diverse cultural and literacy contexts:

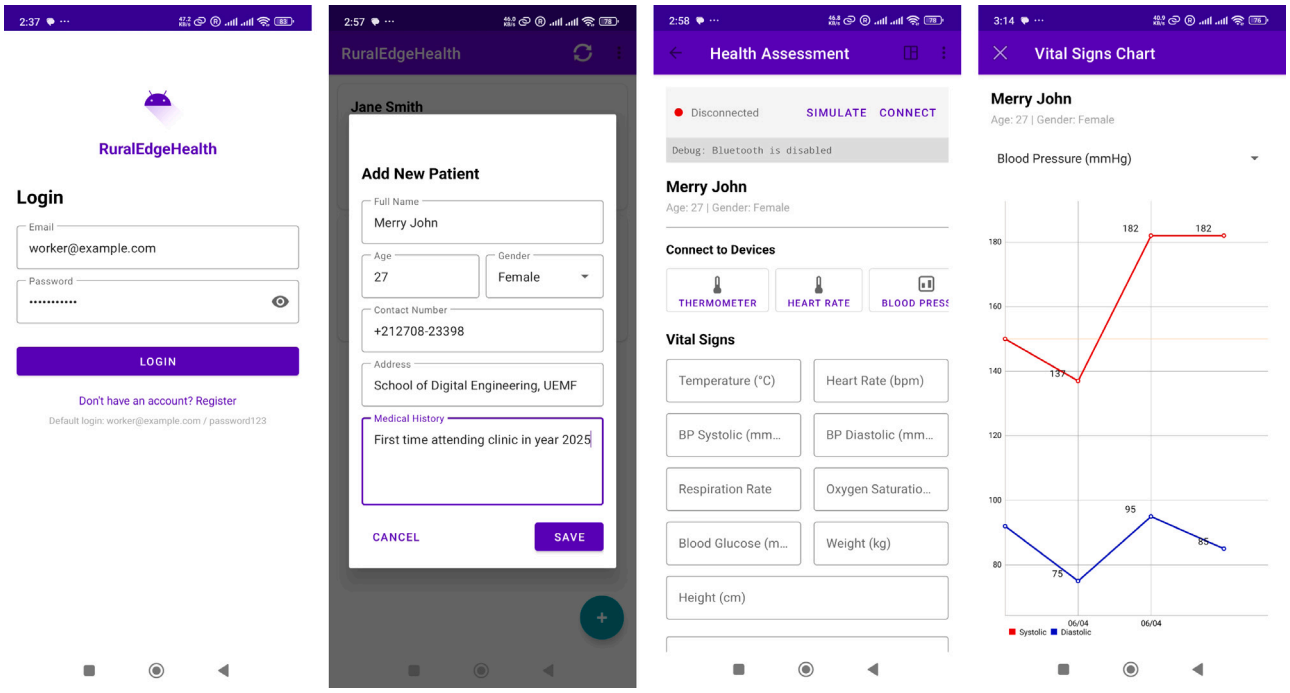
- **Universal Visual Language**: Color cues, iconography, and spatial grouping reduce reliance on text while preserving precision.
- **Progressive Disclosure**: Information is layered to avoid overwhelming novice users while retaining depth for advanced users.
- **Gesture-Based Navigation**: Touch-first navigation reduces interaction overhead on small screens.
- **Offline-First Interaction**: All core interactions function without connectivity, with clear UI feedback for local state and synchronization status.

5. Results

5.1. Model performance comparison

Comprehensive evaluation of the four candidate models revealed strong performance across all tested algorithms. Table 2 summarizes the performance and deployment-relevant characteristics of each approach.

We report results for a health-oriented sensor classification case study; however, the evaluation protocol and performance constraints (offline execution, latency, and battery impact) generalize to other IoT domains.



(a) Authentication interface with accessibility features.

(b) Record/profile creation with offline-capable entry.

(c) Dual-mode data acquisition with BLE integration.

(d) Interactive visualization with trend analysis.

Fig. 8. RuralEdgeHealth core interface workflow demonstrating the user journey from authentication to visualization, emphasizing accessibility and functional clarity for diverse user populations.

Table 3
Class-specific performance metrics for Random Forest model.

Class	Prec.	Rec.	F1	Supp.
Normal	0.99	0.99	0.99	1000
Moderate	0.98	0.97	0.97	700
Critical	0.95	0.97	0.96	300
Weighted Avg	0.98	0.98	0.98	2000

The Random Forest model demonstrated the strongest overall performance, achieving the highest accuracy (98.25%) and F1 score (0.9825) while maintaining competitive inference speed and an acceptable model size for deployment on resource-constrained devices. This trade-off profile motivated its selection for on-device integration.

Class-specific performance analysis indicated strong and balanced performance across all status classes. Table 3 reports the per-class metrics for the Random Forest model.

The confusion matrices for all evaluated models provide detailed insight into classification error patterns, as shown in Fig. 11.

ROC curve analysis further confirmed the strong discriminative capability of the evaluated models, with Random Forest achieving the highest AUC values (Fig. 12).

Feature importance analysis indicated that raw sensor features were the most influential inputs in model decision-making, collectively accounting for 68% of total feature importance, as shown in Fig. 13.

5.2. Cross-device performance benchmarks

Performance evaluation across diverse device configurations confirmed that RuralEdgeHealth maintains acceptable operation on entry-level hardware. Table 4 summarizes cross-device benchmarking results.

5.3. Battery consumption performance analysis

Phase 1 battery performance evaluation demonstrated power efficiency within the deployment targets, as shown in Table 5.

5.4. System usability scale (SUS) evaluation results

Phase 1 usability assessment across diverse user populations provided insights into accessibility design requirements, as shown in Table 6.

Key Finding: The evaluation revealed a **37.5-point SUS gap** between technical users (85.6) and low-literacy users (48.1), quantifying the accessibility challenge and underscoring the importance of inclusive design in offline-capable edge AI systems.

5.5. Task completion performance analysis

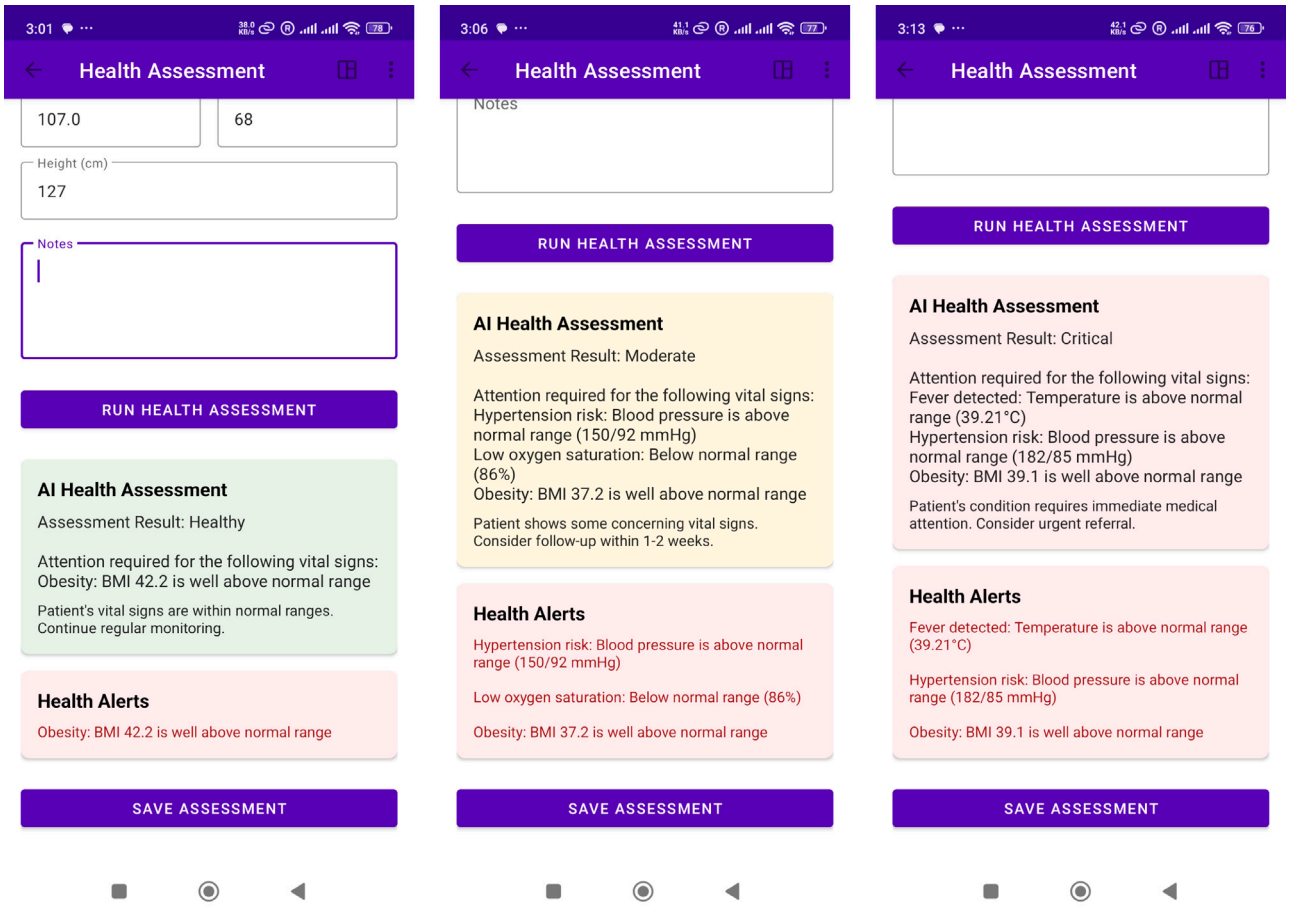
Phase 1 task completion assessment demonstrated robust technical operation across the core workflow, as shown in Table 7.

6. Comparative analysis

This section presents a systematic comparative evaluation of RuralEdgeHealth against representative edge AI IoT systems reported in the recent literature. Although the proposed architecture is evaluated using a health-oriented sensor classification case study, the comparative analysis focuses on system-level capabilities that generalize across intelligent IoT domains.

6.1. Comparison methodology

The comparative analysis considers recent edge AI systems published between 2019 and 2025, selected based on their relevance to edge computing, IoT and sensor integration, and deployment in resource-constrained environments. Rather than emphasizing application-specific outcomes, the comparison adopts an architecture- and deployment-centric perspective, examining capabilities that directly affect real-world feasibility under limited connectivity, energy, and hardware constraints.



(a) Normal classification with standard guidance.

(b) Moderate classification with enhanced recommendations.

(c) Critical classification with immediate alerts.

Fig. 9. AI result interfaces demonstrating adaptive status classification with evidence-based guidance. The system employs color-coded hierarchies and structured presentation to facilitate rapid interpretation across diverse user literacy levels.

Table 4
Cross-device performance benchmarks.

Device model	Processor	RAM	Device tier	Inference (ms)	UI Resp.	Battery
Android Emulator	x86_64 (AVD)	2 GB	Development	89	94%	N/A
Samsung Galaxy A10	Exynos 7884	2 GB	Entry-level	87	87%	3.8%
Xiaomi Redmi Note 12S	MediaTek Helio G96	8 GB	Mid-range	68	99%	3.2%

Table 5
Phase 1 battery consumption analysis by device and usage mode.

Device/Usage scenario	Mean (%/hour)	Range	Sessions (n)	Performance
Samsung Galaxy A10 - Mixed Use	3.8	3.2–4.2	6	Good
Samsung Galaxy A10 - Standby	2.8	2.3–3.1	6	Excellent
Xiaomi Redmi Note 12S - Mixed Use	3.2	2.8–3.6	6	Excellent
Xiaomi Redmi Note 12S - Standby	2.2	1.9–2.7	6	Excellent
Phase 1 overall average	**3.0**	**1.9-4.2**	**24**	Good

Table 8 summarizes a feature-level comparison across the selected systems. The evaluation criteria are derived exclusively from explicitly reported system characteristics in the corresponding publications, ensuring that the comparison reflects verifiable and documented capabilities rather than inferred assumptions.

6.2. Research positioning and contribution

The comparative analysis positions RuralEdgeHealth at the intersection of three deployment-critical challenges that remain largely under-addressed in existing edge AI IoT systems: complete offline operation, compatibility with entry-level mobile hardware, and systematic validation of usability and energy efficiency.

While prior systems demonstrate isolated advances in edge inference, IoT integration, or mobile deployment, none simultaneously combine complete offline inference, support for entry-level Android devices, quantified deployment metrics, manual fallback operation, and explicit usability evaluation for resource-constrained contexts. RuralEdgeHealth addresses this gap through an architecture-level contribution that prioritizes deployability, accessibility, and operational feasibility alongside predictive performance. Its novelty, therefore, lies

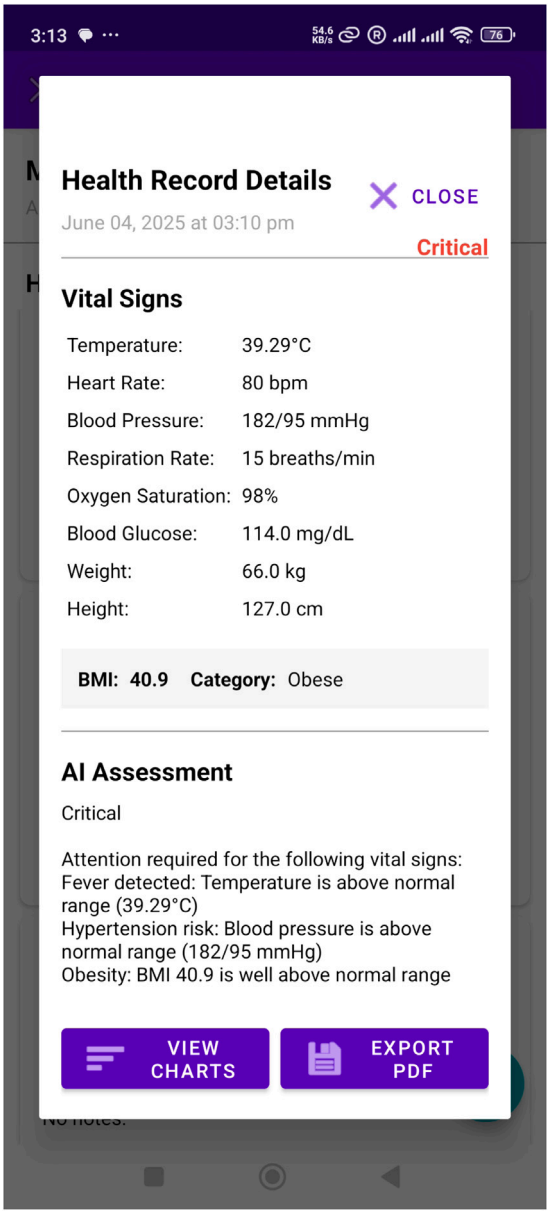


Fig. 10. Record interface showing sensor data, derived metrics, AI assessment outputs, and export functionality for documentation and system integration.

in the integration of these capabilities into a reusable offline-first reference architecture rather than in algorithmic innovation alone

Specifically, the key contributions highlighted by the comparative analysis are:

(1) Complete offline AI inference on entry-level mobile devices without reliance on cloud connectivity.
(2) Systematic quantification of latency, energy consumption, and memory usage to support deployment planning.
(3) Empirical usability evaluation across heterogeneous user groups, informing accessibility-aware interface design.
(4) Architecture-level optimization explicitly targeting resource-constrained and low-connectivity environments.

These characteristics position RuralEdgeHealth as a generalizable edge AI IoT reference architecture, designed not only for algorithmic performance but also for equitable, real-world deployment across diverse intelligent IoT application domains.

Table 6
Phase 1 SUS evaluation results by user group.

User group	n	Mean SUS	SD	Range	Usability level
Technical users	4	85.6	6.2	78–93	Excellent
General users	4	65.6	11.8	50–77	Below average
Low-literacy users	4	48.1	13.7	30–62	Poor
Overall Phase 1	**12**	**66.4**	**18.9**	**30–93**	**Below average**

Table 7
Phase 1 task completion analysis (n = 12 participants).

Task function	Success rate	Completion	Performance level
Data registration	100.0%	12/12	Excellent
Sensor data entry	100.0%	12/12	Excellent
AI assessment	100.0%	12/12	Excellent
Data export	91.7%	11/12	Good
BLE device pairing	75.0%	9/12	Requires enhancement
Overall performance	**93.3%**	**56/60**	**Good**

7. Discussion

7.1. Principal findings and technical validation

This study demonstrates that AI-driven IoT sensor classification can achieve high accuracy and low latency on entry-level mobile devices without reliance on continuous network connectivity. Achieving 98.25% classification accuracy with sub-100 ms inference latency on devices equipped with only 2 GB of RAM directly challenges prevailing assumptions that intelligent IoT applications require persistent cloud access or high-performance hardware. In addition, the quantified energy consumption of less than 4.0% per hour supports practical full-day field deployment, addressing a critical gap in deployment-oriented performance reporting within the edge AI literature.

Collectively, these results provide empirical evidence that offline-first mobile edge AI is technically viable under strict hardware, energy, and connectivity constraints. Rather than optimizing for peak performance under ideal conditions, the proposed architecture prioritizes robustness and accessibility, aligning system design with real-world operating environments common in underserved and infrastructure-limited contexts.

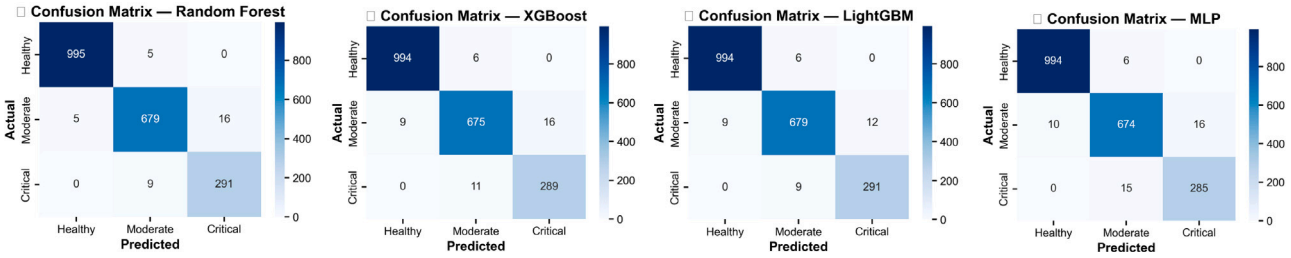
7.2. Accessibility and usability insights

The systematic usability evaluation revealed a substantial 37.5-point System Usability Scale (SUS) gap between technical users (85.6) and low-literacy users (48.1). This finding represents one of the first quantitative assessments of accessibility disparities in mobile edge AI IoT systems and highlights an often-overlooked limitation in existing deployments: technical correctness does not imply universal usability.

The observed disparity indicates that interfaces optimized for technically trained users may inadvertently exclude the populations most intended to benefit from offline-capable IoT systems. These results emphasize that accessibility must be treated as a first-class design constraint rather than a secondary consideration. From an architectural perspective, the findings motivate the inclusion of adaptive interfaces, simplified interaction workflows, and context-aware guidance mechanisms as integral components of equitable edge AI systems.

7.3. Implications for digital equity and inclusive infrastructure

By enabling high-performance AI inference without continuous connectivity or high-end hardware, RuralEdgeHealth addresses structural barriers affecting an estimated 2.6 billion people worldwide who lack reliable internet access. The results demonstrate that intelligent IoT



(a) Random Forest (b) XGBoost (c) LightGBM (d) Multi-Layer Perceptron

Fig. 11. Confusion matrices for all evaluated models showing classification performance across status classes.

Table 8
Feature comparison of edge AI IoT systems.

	Pathinarupothi (2019)	Rathi (2021)	Cruz (2024)	Wang (2025)	Qi (2025)	RuralEdgeHealth (2025)
Basic capabilities						
Edge processing	✓	✓	✓	✓	✓	✓
IoT/Sensor integration	✓	✓	✓	×	✓	✓
Mobile platform support	✓	✓	×	×	✓	✓
Advanced features						
Complete offline AI	×	✓	×	×	×	✓
Entry-level device support	×	×	×	×	×	✓
Quantified performance metrics	×	×	×	×	×	✓
Manual input fallback	×	×	×	×	×	✓
Validation						
Battery consumption analysis	×	×	×	×	×	✓
Formal usability testing	×	×	×	×	×	✓
Deployment						
No connectivity required	×	✓	×	×	×	✓
Explicit resource-constrained design	×	×	×	×	×	✓

services need not be restricted to well-connected or resource-rich environments, provided that system architectures are explicitly designed around accessibility constraints.

Beyond the specific case study evaluated in this work, the findings support a broader implication: digital equity in intelligent infrastructure requires offline-first, resource-aware architectural paradigms. The demonstrated feasibility of such systems contributes technical evidence toward inclusive innovation goals aligned with global digital development and smart infrastructure initiatives.

7.4. Comparison with state-of-the-art design choices

The selection of a Random Forest model over deep learning alternatives reflects a deliberate design trade-off informed by deployment constraints rather than algorithmic novelty. While deep neural networks often dominate performance benchmarks under unconstrained conditions, their large memory footprint, high inference latency, and elevated energy consumption limit their practicality on entry-level devices.

In contrast, tree-based models provide a favorable balance between accuracy, computational efficiency, interpretability, and deployment feasibility. The comparative analysis illustrates that, for offline mobile edge AI systems operating under strict resource limits, classical machine learning approaches may offer superior end-to-end performance compared to more complex deep learning architectures. This finding reinforces the importance of deployment-aware model selection in edge AI research.

7.5. Generalizability and domain adaptation

Although the proposed architecture is demonstrated using a health-oriented sensor classification scenario, its modular five-layer design supports straightforward adaptation to other IoT application domains. Domain transfer primarily requires substitution of sensor interfaces, feature extraction pipelines, and trained ONNX models, without modification to the core architectural components.

Potential application domains include environmental monitoring, agricultural sensing, industrial condition monitoring, and wearable-based systems. This adaptability confirms that RuralEdgeHealth functions as a reusable reference architecture rather than a domain-specific solution, enabling replication and extension across diverse intelligent IoT contexts.

7.6. Threats to validity

7.6.1. Internal validity

Synthetic Dataset Usage: The use of domain-informed synthetic data enabled controlled experimentation and reproducible evaluation but cannot fully capture real-world variability. Although the dataset preserves plausible relationships among sensor, contextual, and device-level variables and includes controlled measurement noise, it remains a feasibility-oriented approximation rather than a substitute for real deployment data. Future validation will therefore incorporate real BLE sensor streams and real-world datasets where available.

Controlled Evaluation Conditions: Performance metrics were obtained under standardized laboratory settings to support repeatable comparison across representative devices. However, real deployments may introduce additional variability arising from BLE instability, calibration drift, sensor degradation, environmental noise, data missingness, and heterogeneous user behavior. These factors may affect both predictive performance and usability and therefore, require dedicated field validation.

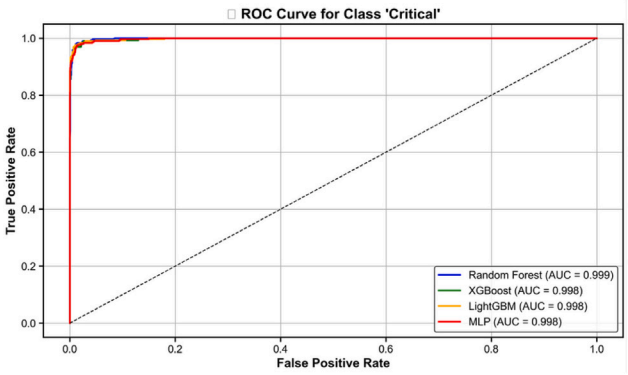


Fig. 12. ROC curves comparing all models, showing excellent discriminative capability across status classes, with Random Forest achieving the highest AUC values.

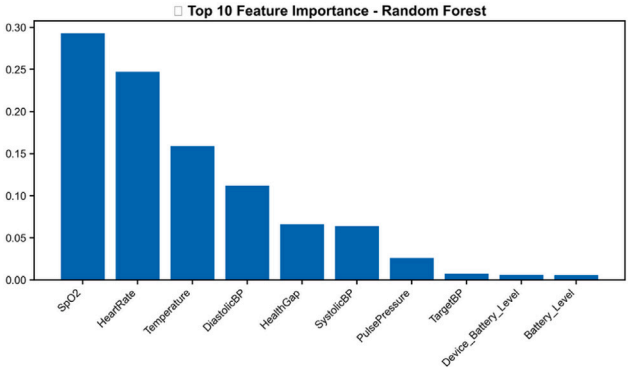


Fig. 13. Feature importance analysis for the Random Forest model showing the relative contribution of each feature and contextual factor to classification decisions.

7.6.2. External validity

Sample Size Constraints: The Phase 1 usability evaluation involved a limited number of participants (n = 12). While sufficient for exploratory insights, larger-scale studies are required to support statistically robust generalization.

Device Diversity: Validation was conducted on a small number of representative Android devices. The global Android ecosystem includes a wide range of hardware configurations that may exhibit different performance characteristics.

7.6.3. Construct validity

Classification Taxonomy: The three-class Normal/Moderate/Critical labeling scheme is used to demonstrate architectural capability rather than domain-specific diagnostic precision. Domain-specific deployments would require task definitions aligned with established standards and stakeholder requirements.

7.7. Limitations

Several limitations constrain the scope of the present study. First, the usability evaluation represents an exploratory Phase 1 assessment rather than a comprehensive user study and should therefore be interpreted as providing early design guidance rather than definitive population-level conclusions. Second, the controlled experimental environment was intentionally used to support reproducible technical feasibility assessment, but it does not fully reflect long-term field deployment conditions. Third, reliance on synthetic data supports architectural and computational feasibility analysis but limits conclusions regarding real-world data heterogeneity, calibration drift, and operational variability. Finally, the evaluation period does not capture long-term system reliability, interface adaptation over time, or sustained real-world user behavior.

Despite these limitations, the presented results provide a strong technical foundation for future large-scale validation and domain-specific deployment studies.

8. Future work

Future research priorities emerge directly from Phase 1 findings and limitations:

Immediate Development Priorities: Accessibility enhancement through comprehensive interface redesign targeting low-literacy users, including visual simplification, cultural adaptation, and intuitive navigation workflows.

Validation Studies: Large-scale usability validation (e.g., n $\geqslant$ 50) across more diverse technical, non-technical, and low-literacy user groups; domain-specific validation using real-world datasets and live BLE sensor streams under appropriate oversight; and pilot field deployment studies in practical operating environments to examine distribution shift, calibration drift, connectivity instability, and long-term usability

Technical Enhancements: Hybrid connectivity architecture enabling cloud synchronization when available while maintaining offline-first principles; federated learning integration for privacy-preserving model improvement; multi-language support for interface localization.

Advanced Research Directions: Cross-platform implementation exploration (Flutter/React Native for iOS support); advanced sensor integration beyond BLE (ANT+, Zigbee, WiFi-Direct); integration APIs for existing information systems.

9. Conclusion

This study establishes the technical feasibility of deploying offline-first edge AI frameworks on widely accessible, resource-constrained Android smartphones. The proposed reference architecture, RuralEdgeHealth, demonstrated high performance by achieving 98.25% classification accuracy and sub-100 ms inference latency using a compact 3.5 MB ONNX model on entry-level devices with only 2 GB RAM. These findings challenge prevailing assumptions about the infrastructure required for sophisticated AI applications and affirm the viability of local inference in disconnected environments.

Cross-device validation confirmed consistent system behavior across diverse hardware configurations, supporting scalable deployment in low-resource settings. Usability testing across multiple user populations revealed critical accessibility disparities—particularly the 37.5-point SUS gap between technical and non-technical users—highlighting the need for inclusive design approaches in edge AI systems. The integration of BLE-based sensor data acquisition with fallback manual input ensures robustness even in sensor-degraded conditions. By combining a modular five-layer reference architecture, optimized machine learning deployment, and evidence-based interface design, RuralEdgeHealth contributes a reproducible and open-source benchmark for AI-enabled IoT applications in underserved contexts. The offline-first design offers a practical pathway to extend intelligent IoT services to populations lacking reliable internet access. At the same time, the present findings should be interpreted within the scope of a Phase 1 feasibility study. The usability results provide exploratory evidence of accessibility disparities that can guide interface redesign, while the synthetic-data-based evaluation establishes technical feasibility rather than real-world domain validity. Future work will therefore extend this foundation through larger and more diverse usability studies, validation with real sensor data, and field-based deployment testing.

This work presents a reference architecture contribution demonstrating technical feasibility, not a validated system for critical deployments. Future work will focus on domain-specific validation, accessibility enhancement through human-centered interface refinement, and optional hybrid cloud synchronization while maintaining core offline-first principles.

CRediT authorship contribution statement

Abubakar Wakili: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Data curation, Conceptualization. **Sara Bakkali:** Writing – review & editing, Conceptualization, Methodology, Project administration, Supervision.

Declaration of competing interest

The authors declare no conflict of interest regarding the publication of this research.

Data availability

The dataset and source code supporting the findings of this study are publicly available via the open-source repository provided in the accompanying Software Impacts article.

References

[1] Maita KC, Maniaci MJ, Haider CR, Avila FR, Torres-Guzman RA, Borna S, Lunde JJ, Coffey JD, Demaerschalk BM, Forte AJ. The impact of digital health solutions on bridging the health care gap in rural areas: A scoping review. Perm J 2024;28:130–43. http://dx.doi.org/10.7812/TPP/23.134.

[2] Ahmed MM, Okesanya OJ, Olaleke NO, Adigun OA, Adebayo UO, Oso TA, Eshun G, Lucero-Prisno III DE. Integrating digital health innovations to achieve universal health coverage: Promoting health outcomes and quality through global public health equity. Heal 2025;13:1060. http://dx.doi.org/10.3390/healthcare13091060.

[3] Yi S, Yam ELY, Cheruvettolil K, Linos E, Gupta A, Palaniappan L, Rajeshuni N, Vaska KG, Schulman K, Eggleston KN. Perspectives of digital health innovations in low- and Middle-Income health care systems from south and Southeast Asia. J Med Internet Res 2024;26:e57612. http://dx.doi.org/10.2196/57612.

[4] International Telecommunication Union. Facts and figures 2024: global connectivity report. ITU Media Centre; 2024, https://www.itu.int/en/mediacentre/Pages/PR-2024-11-27-facts-and-figures.aspx.

[5] Badr J, Motulsky A, Denis JL. Digital health technologies and inequalities: A scoping review of potential impacts and policy recommendations. Health Policy 2024;105122. http://dx.doi.org/10.1016/j.healthpol.2024.105122.

[6] Wakili A, Bakkali S. Internet of Things in healthcare: An adaptive ethical framework for IoT in digital health. Clin EHealth 2024;7:92–105. http://dx.doi.org/10.1016/j.ceh.2024.07.001.

[7] Saeed SA, Campbell R. Disparities in health care and the digital divide. Curr Psychiatry Rep 2021;23(9):61. http://dx.doi.org/10.1007/s11920-021-01274-4.

[8] United Nations. Health - sustainable development goal 3. 2023, [Online]. Available: https://www.un.org/sustainabledevelopment/health/. [Accessed 6 June 2025].

[9] World Health Organization and World Bank. Tracking universal health coverage: 2023 global monitoring report. Geneva: WHO; 2023, [Online]. Available: https://www.who.int/publications/i/item/9789240080379.

[10] Amin SU, Hossain MS. Edge intelligence and Internet of Things in healthcare: A survey. IEEE Access 2021;9:45–59. http://dx.doi.org/10.1109/ACCESS.2020.3045115.

[11] Hartmann M, Hashmi US, Imran A. Edge computing in smart health care systems: Review, challenges, and research directions. Trans Emerg Tel Tech 2022;33:e3710. http://dx.doi.org/10.1002/ett.3710.

[12] Rancea A, Anghel I, Cioara T. Edge computing in healthcare: Innovations, opportunities, and challenges. Future Internet 2024;16(9):329. http://dx.doi.org/10.3390/fi16090329.

[13] Yaraziz MS, Safa NS, Azad MA. Edge computing in IoT for smart healthcare. J Ambient Intell Smart Environ 2024;16:409–38. http://dx.doi.org/10.3233/AIS-230009.

[14] Gao X, He P, Zhou Y, Qin X. A smart healthcare system for remote areas based on the Edge–Cloud continuum. Electronics 2024;13:4152. http://dx.doi.org/10.3390/electronics13214152.

[15] Wang X, Li Y, Chen Z. Optimizing edge AI: A comprehensive survey on data, model, and system strategies. 2025, arXiv preprint Online. Available: https://arxiv.org/html/2501.03265v1.

[16] Tawalbeh LA, Muheidat F, Tawalbeh M, Quwaider M, AA Abd El-Latif. Edge enabled IoT system model for secure healthcare. Measurement 2022;191:110792. http://dx.doi.org/10.1016/j.measurement.2022.110792.

[17] WA Cruz Castañeda, P Bertemes Filho. Improvement of an Edge-IoT architecture driven by artificial intelligence for Smart-Health chronic disease management. Sensors 2024;24:7965. http://dx.doi.org/10.3390/s24247965.

[18] Alabdulatif A, Khalil I, Yi X, Guizani M. Secure edge of things for smart healthcare surveillance framework. IEEE Access 2019;7:31010–21. http://dx.doi.org/10.1109/ACCESS.2019.2899323.

[19] Rathi V, Kumar P, Singh M, Malviya R, Rathore G. An edge AI-enabled IoT healthcare monitoring system for smart cities. Comput Electr Eng 2021;96:107524. http://dx.doi.org/10.1016/j.compeleceng.2021.107524.

[20] Lamem MFH, Sahid MI, Ahmed A. Artificial intelligence for access to primary healthcare in rural settings. J Med Surg Public Health 2025;5:100173. http://dx.doi.org/10.1016/j.glmedi.2024.100173.

[21] KM Saif-Ur-Rahman, Islam MS, Alaboson J, Ola O, Hasan I, Islam N, et al. Artificial intelligence and digital health in improving primary health care service delivery in LMICs: A systematic review. J Evid Based Med 2023;16:303–20. http://dx.doi.org/10.1111/jebm.12547.

[22] Dias J. Wearable health devices—Vital sign monitoring, systems and technologies. Sensors 2018;18:2414. http://dx.doi.org/10.3390/s18082414.

[23] Surianarayanan C, Lawrence JJ, Chelliah PR, Prakash E, Hewage C. A survey on optimization techniques for edge artificial intelligence (AI). Sensors 2023;23:1279. http://dx.doi.org/10.3390/s23031279.

[24] Pathinarupothi RK, Durga P, Rangan ES. IoT-Based smart edge for global health: Remote monitoring with severity detection and alerts transmission. IEEE Internet Things J 2019;6:2449–62. http://dx.doi.org/10.1109/JIOT.2018.2870068.

[25] Izhar M, Naqvi SAA, Ahmed A, Abdullah S, Alturki N, Jamel L. Enhancing healthcare efficacy through IoT-Edge fusion: A novel approach for smart health monitoring and diagnosis. IEEE Access 2023;11:136456–67. http://dx.doi.org/10.1109/ACCESS.2023.3337092.

[26] Talpur MSH, Alam Bhuiyan Z, Wang G. Energy-efficient healthcare monitoring with smartphones and IoT technologies. Int J High Perform Comput Netw 2022;8:186–94. http://dx.doi.org/10.1504/IJHPCN.2015.070019.

[27] Mishra AD, Thakral B, Jijja A, Sharma N. Real-time vital signs monitoring and data management using a Low-Cost IoT-based health monitoring system. J Health Manag 2024. http://dx.doi.org/10.1177/09720634241246926.

[28] HA El Zouka, Hosni MM. Secure IoT communications for smart healthcare monitoring system. Internet Things 2021;13:100036. http://dx.doi.org/10.1016/j.iot.2019.01.003.

[29] A Alzu'Bi, Aloqaily M, Guizani M, A Al-Fuqaha. A review of privacy and security of edge computing in smart healthcare systems: issues, challenges, and research directions. Tsinghua Sci Technol 2024;29:1152–80. http://dx.doi.org/10.26599/TST.2023.9010080.

[30] Okwu EA, Tartibu LK, Okwu FO. Artificial intelligence for strengthening healthcare systems in low- and middle-income countries: a systematic scoping review. Npj Digit Med 2022;5(1):162. http://dx.doi.org/10.1038/s41746-022-00700-y.

[31] HealthEdgeAI: GAI and XAI based healthcare system for sustainable edge AI and cloud computing environments. Concurr Comput Pr Exp 2025;37:e70057. http://dx.doi.org/10.1002/cpe.70057.

[32] Hennebelle A, Materwala H, Ismail L. HealthEdge: a machine learning-based smart healthcare framework for prediction of type 2 diabetes in an integrated IoT, edge, and cloud computing system. Procedia Comput Sci 2023;220:331–8. http://dx.doi.org/10.1016/j.procs.2023.03.043.

[33] Giuffrè M, Shung DL. Harnessing the power of synthetic data in healthcare: innovation, application, and privacy. Npj Digit Med 2023;6:186. http://dx.doi.org/10.1038/s41746-023-00927-3.

[34] Loni M, Poursalim F, Asadi M, et al. A review on generative AI models for synthetic medical text, time series, and longitudinal data. Npj Digit Med 2025;8:281. http://dx.doi.org/10.1038/s41746-024-01409-w.

[35] Boraschi D, et al. Governing synthetic data in medical research: the time is now. Lancet Digit Health 2025;7(4):e233–4. http://dx.doi.org/10.1016/j.landig.2025.01.012.